
Automated Literature-Based Target Ranking Reveals PAI-1 (SERPINE1) as a Priority in IPF Aging–Fibrosis Network

GPT-5.2, Gemini 3 Pro and Perplexity Pro



Abstract

This study employed a RAG-driven pipeline to prioritize PAI-1 (SERPINE1) as a key therapeutic target for IPF, evaluating candidates based on efficacy, reproducibility, and mechanism (C1–C3). Validation confirmed that PAI-1 inhibition effectively reduces fibrosis markers and p53/p21-mediated senescence. By integrating network analysis with structure-based modeling, the research proposed novel peptide drug candidates, successfully bridging the gap between target discovery and drug design.

1 Introduction

Idiopathic pulmonary fibrosis (IPF) is a representative chronic lung disease characterized by progressive fibrosis of the lung parenchyma and deterioration of respiratory function, occurring without a clearly identified cause. Pathologically, it is marked by alveolar structural collapse, expansion of fibrotic lesions, increased extracellular matrix (ECM) accumulation, and sustained activation of fibroblasts/myofibroblasts. Clinically, it is often difficult to sufficiently suppress progression, necessitating strategies for identifying new therapeutic targets.

Recently, cellular senescence has been discussed as a key axis explaining IPF progression. When the p53–p21 axis is activated by injury or stress, cell cycle arrest is induced, and senescent cells can alter the surrounding microenvironment, amplifying inflammatory and fibrotic responses. Particularly in ATII cells, aging and functional decline may be linked to failed alveolar regeneration, and a model is proposed where SASP signaling reinforces fibroblast activity and ECM accumulation, thereby cementing a fibrotic phenotype.

Therefore, identifying key regulatory factors that can be targeted within the translational cascade—where upstream damage/stress signals translate into downstream fibrosis execution (e.g., ECM accumulation) via aging-related reprogramming (e.g., p53/p21)—can enhance target discovery efficiency. However, literature-based discovery incurs high manual integration costs due to candidate overload, heterogeneity of evidence formats, and evidence dispersion. This study derives priority verification candidate targets by structuring evidence through a literature-based RAG automated evaluation pipeline and consistently assessing it against C1/C2/C3 criteria. This paper presents the assessment distribution and final selection results in Figures 1–2, the PAI-1 evidence and integrated model in Figures 3–7 and Figure 9, the structure-based candidate proposal results in Figure 8 and Table 1, and the follow-up validation roadmap and limitation/complementation plan in Tables 2–3.

2 Methods

2.1 Literature-based RAG automated evaluation pipeline

This study established a literature-based RAG automated evaluation pipeline to prioritize candidate targets within the aging–fibrosis network. The pipeline follows a five-stage process: (1) literature collection using keywords such as IPF, senescence, TGF- β , and PAI-1; (2) information extraction of target molecules, experimental models, intervention methods, and outcome variables; (3) structuring and normalization of data into a defined schema (Target $\times$ Evidence $\times$ Model $\times$ Intervention $\times$ Readout $\times$ Direction/Value); (4) automatic evaluation against predefined criteria; and (5) final output of screening summaries. Extracted outcomes focus on fibrotic indicators like collagen quantification, α -SMA, and fibrotic scores. Information is conservatively classified as UNCERTAIN if literature evidence is ambiguous.

2.2 Automated evaluation criteria (C1/C2/C3)

Targets are assessed along three core axes to ensure translational relevance. C1 (Efficacy) evaluates whether direct modulation of the target via inhibitors or genetic knockdown significantly reduces fibrosis markers like collagen or ECM accumulation. C2 (Reproducibility) assesses whether results are replicated across independent research groups and different experimental models. C3 (Mechanistic connectivity) determines if the target occupies a plausible position within the pathway axis—linking upstream damage signals to aging-related reprogramming (p53/p21) and downstream fibrotic execution. Final selection prioritizes C1; targets with supporting C2 or C3 evidence are classified as KEEP (priority verification), while those with insufficient evidence are labeled AMBIGUOUS.

2.3 Presentation of Results (Tables/Figures)

The research findings are structured within a nine-page limit, encompassing automated assessment distributions (Figures 1–2), PAI-1 efficacy and mechanistic evidence (Figures 3–5), and integrated network models (Figures 6–7, 9). Structure-based peptide candidate proposals and the subsequent validation roadmap are summarized in Figure 8 and Tables 1–3.

3 Results

3.1 Summary of automated evaluation and rationale for selecting PAI-1

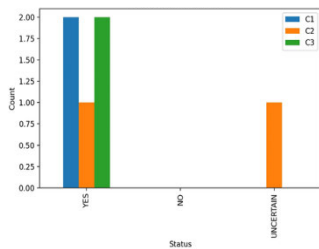


Figure 1: Distribution of C1/C2/C3 Automated Assessment Results.

We assessed candidate targets extracted and normalized from the input literature ($n = 27$) using C1 (Efficacy), C2 (Reproducibility), and C3 (Mechanistic connectivity) as YES/NO/UNCERTAIN. YES judgments were relatively frequent for C1 and C3, while C2 was often UNCERTAIN due to insufficient cross-study validation.

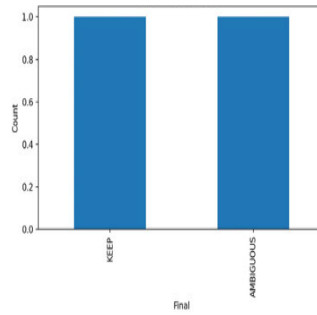


Figure 2: Summary of final screening results.

Candidates were classified as KEEP (priority validation) or AMBIGUOUS (additional evidence required) by integrating C1/C2/C3 assessments. KEEP was defined as candidates with minimal strong evidence on a core axis (C1 or C3), while acknowledging remaining gaps as addressable in subsequent phases. AMBIGUOUS was defined as candidates with insufficient, conflicting, or unclear evidence on the core axis. Under these criteria, PAI-1 (SERPINE1) was designated as a priority validation target and selected as the central candidate for subsequent integration and drug-candidate proposal stages

3.2 PAI-1 (SERPINE1) C3 (Mechanism) Basis: Feedback Coupling with the p53/p21 Axis

Direct inhibition of PAI-1 has been reported to reduce fibrosis markers in vivo. In the study by Omori et al. (2016), administration of the PAI-1 inhibitor TM5275 under fibrosis-inducing conditions resulted in reduced collagen quantification (hydroxyproline). Collagen accumulation elevated under induction conditions decreased toward control levels upon TM5275 treatment (Figures 3–4). These results provide an efficacy anchor satisfying the C1 criterion of “direct intervention leading to reduction in fibrosis markers.”

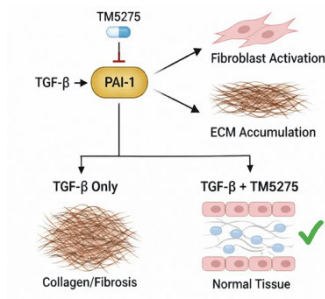


Figure 3: Schematic of the antifibrotic mechanism of PAI-1 inhibition (TM5275).

This diagram illustrates the pathway linking PAI-1 to fibroblast activation and ECM accumulation (collagen accumulation) under TGF-β-induced conditions, summarizing the concept that PAI-1 inhibition by TM5275 leads to attenuation of the fibrotic phenotype.

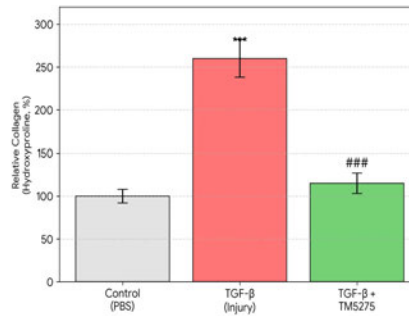


Figure 4: Fibrosis inhibition (C1) demonstrated by reduced collagen quantification (hydroxyproline) following TM5275 administration.

Comparison of hydroxyproline-based relative collagen content (%) among the control group (PBS), TGF- β -induced group, and TGF- β +TM5275-treated group demonstrates that the increased collagen content in the TGF- β -induced group is significantly reduced upon TM5275 administration (statistical significance indicated by symbols within the graph).

3.3 C3 mechanistic connectivity for PAI-1: linkage with p53/p21 axis

To assess whether PAI-1 occupies a mechanistically plausible position within the aging–fibrosis axis, evidence for its linkage with the p53/p21 axis is summarized in C3. Figure 5a conceptually summarizes, based on the literature by Jiang et al. (2017), that PAI-1 could be incorporated into a feedback structure associated with p53 stabilization. Figure 5b presents results showing that relative expression of p53 protein and p21 protein decreased compared to the control group when PAI-1 was suppressed by siRNA. Therefore, this study classified PAI-1 as a candidate with C3 (mechanism connectivity) evidence, as it possesses intervention-based evidence: “PAI-1 regulation (intervention) $\rightarrow$ p53/p21 changes (outcome)” (Figure 5a–5b).

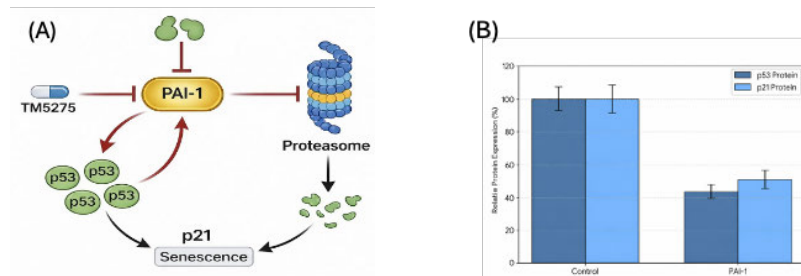


Figure 5: Mechanistic and quantitative evidence that PAI-1 inhibition attenuates the p53–p21 senescence axis.

(A) Schematic model: PAI-1 is associated with p53 stabilization/accumulation, thereby reinforcing p21-mediated cellular senescence; pharmacological inhibition of PAI-1 (TM5275) is summarized as blocking this axis and weakening senescence-locking signaling.

(B) Quantification: Under PAI-1 inhibition, p53 and p21 protein levels are reduced compared with the control condition (bars indicate mean; error bars indicate variability).

3.4 Network perspective integration: Summarizing PAI-1 as the connecting point of the upper–core–lower axis

This study integrated the evidence for PAI-1 (SERPINE1) derived from self-assessment within the aging–

fibrosis axis framework. Figure 6 summarizes this by connecting PAI-1 at the center with upstream inducing signals (TGF- β /NOX4), senescence arrest (p53/p21), the inflammation/paracrine axis (IL-11/SASP), and the final fibrotic outcome (fibroblast activation/ECM accumulation) into a single schematic. Figure 7 contrasts regeneration (cell cycle re-entry) at low PAI-1 levels with aging arrest/fibrosis stasis at high PAI-1 levels, providing a conceptual model suggesting that PAI-1 inhibition (TM5275) enables a shift toward recovery. Therefore, this paper proposes PAI-1 as a candidate intervention point where the aging-related axis and the fibrosis execution axis intersect, suggesting it as a priority candidate target for validation (Figures 6–7).

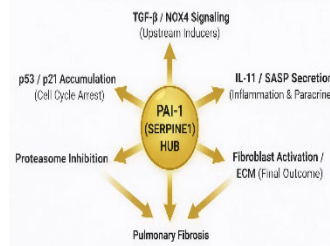


Figure 6: Summary of the PAI-1 (SERPINE1)-centered integrated network.

Based on literature-based evidence, connections between PAI-1 and its upstream inducing signals (TGF- β /NOX4), aging-related fixation (p53/p21), inflammation/paracrine (IL-11/SASP), and final fibrotic outcomes (fibroblast activation/ECM accumulation) are schematically summarized.

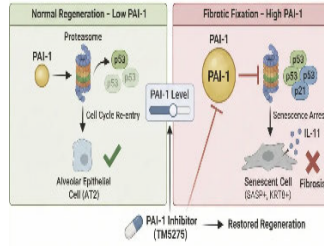


Figure 7: Conceptual diagram of the regeneration versus fibrosis/senescence model and its reversal upon inhibition at different PAI-1 levels.

It contrasts cell cycle re-entry/regeneration under low PAI-1 levels with aging arrest and fibrotic fixation under high levels, conceptually summarizing that PAI-1 inhibition (TM5275) could be linked to a reversal toward recovery.

3.5 Structure – guided peptide candidate proposal targeting PAI-1

This study first identified PAI-1 (SERPINE1) as a priority candidate target through literature-based automated screening. To progress beyond target identification to the “drug candidate proposal” stage, it proposed peptide candidates targeting the PAI-1 binding pocket and evaluated their structure-based binding potential (predicted structure and docking). The small-molecule inhibitor (TM5275) reported in the literature was utilized in Section 3.1 based on its *in vivo* efficacy (C1) evidence for PAI-1 inhibition. This section presents computational binding affinity evidence for the novel peptide candidates.

Figure 8 presents the predicted complex structure (8A) showing the representative pose of the proposed peptide candidate binding to the pocket region on the surface of the PAI-1 protein, along with a view (8B) magnifying the local interactions around the pocket. Table 2 summarizes the docking scores, confidence scores, and the template PDB (e.g., 1DB2) used for the top candidates. These structure-based results should be interpreted as supplementary evidence for candidate prioritization and binding plausibility for subsequent validation, not as confirmation of experimental binding affinity/inhibitory activity.

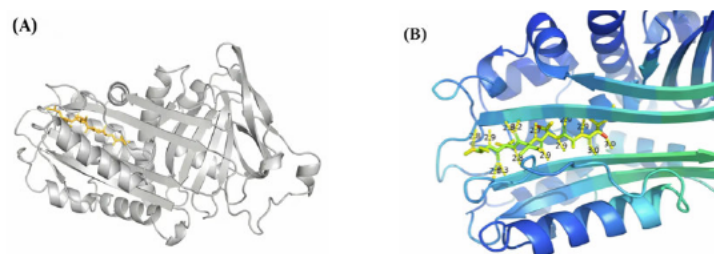


Figure 8: Structure-guided peptide candidate proposal targeting PAI-1 (SERPINE1).

(A) Predicted protein-peptide complex structure showing the representative binding pose of the proposed peptide candidate. (B) Enlarged view of the binding pocket summarizing pocket receptivity and local interactions (e.g., contacting residues/hydrogen bond-forming sites). This figure presents computational binding potential and does not imply experimental confirmation.

Table 1: Proposed peptide candidate targeting PAI-1 (SERPINE1) and supporting structure/docking evidence.

Evaluation Criteria	Numerical (Rank 1)	Interpretation and Scientific Significance
Docking Score	-406.18	Indicates very strong binding energy (generally favorable if below -200).
Confidence Score	0.9941	Predicted complex structure confidence >99% (≥ 0.7 indicates high confidence).

Docking score and confidence metrics were used for ranking within the same computational run and do not imply absolute binding affinity.

4 Discussion

We developed a literature-based RAG automated evaluation pipeline to structure and assess targets in the IPF aging-fibrosis axis using consistent criteria (C1 efficacy, C2 reproducibility, C3 mechanistic connectivity). PAI-1 (SERPINE1) emerged as a priority target based on (i) in vivo efficacy evidence showing reduced hydroxyproline/collagen markers upon TM5275 administration (C1) and (ii) mechanistic connectivity evidence showing decreased p53/p21 expression following PAI-1 inhibition via siRNA (C3). We further integrate PAI-1 within a network model linking upstream induction (TGF- β /NOX4), senescence fixation (p53/p21), paracrine inflammation/SASP, and downstream fibrosis execution (fibroblast activation/ECM accumulation).

Structure-based peptide proposals serve as qualitative supporting evidence suggesting binding plausibility, not confirmation of binding affinity or inhibitory activity. Subsequent validation should follow staged experiments (Table 2), including replication of in vivo efficacy, evaluation of epithelial senescence mechanisms, assessment of epithelial-to-fibroblast crosstalk, and establishment of therapeutic windows. Because PAI-1 participates in physiological fibrinolysis/thrombosis pathways, dose-response and safety monitoring are essential.

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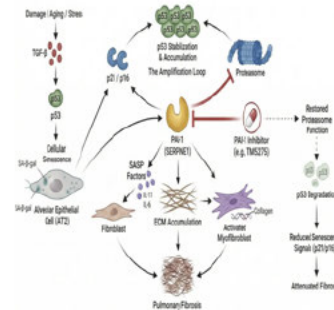


Figure 9: Integrated model of the aging-fibrosis axis through PAI-1 inhibition.

This diagram integrates pathways linking TGF- β -p53-p21($\pm$ p16)-based senescence arrest and SASP, fibroblast activation, and ECM accumulation in injury/aging stress, with PAI-1 at the center. It conceptually summarizes how PAI-1 inhibition (e.g., TM5275) can be linked to attenuating signaling along this axis.

Table 2: Follow-up experimental roadmap for PAI-1 priority validation

Step	Purpose	Model/Setting	Intervention	Primary Readouts	Success Criteria
1	In vivo efficacy re-evaluation (C1)	IPF/pulmonary fibrosis animal model (including therapeutic intervention after disease induction)	PAI-1 inhibition (low molecular weight: TM5275, etc.) or genetic inhibition (KD/KO) (if possible)	collagen/hydroxyproline, α -SMA, fibrotic score (histology), ECM indicators + staining (e.g., Masson), pulmonary functional indicators	Reduction in fibrosis indicators; improvement
2	Validation of epithelial aging axis (C3 enhancement)	ATII or epithelial cell-centered stress/aging induction model	PAI-1 inhibition (siRNA/drug)	p53, p21($\pm$ p16, senescence marker), SASP panel (inflammatory/fibrotic secretory factors)	Weakening of p53/p21 axis; reduced SASP; improved cell cycle recovery signals (e.g., ppRb)
3	Verification of epithelial-to-fibroblast transition (crosstalk)	Epithelial-fibroblast co-culture or conditioned medium	PAI-1 inhibition in epithelial cells followed by conditioned media measurement of fibroblast response	fibroblast α -SMA, collagen, ECM program (transcription/protein), contraction/motility and other functional indicators	Inhibition of fibroblast activity and ECM program (epithelial regulation $\rightarrow$ weakened fibrosis execution)

4	Safety and therapeutic window confirmation / schedule optimization	Repeated administration/dose-response (including toxicity; pharmacokinetics/pharmacodynamics)	PAI-1 inhibitor dose escalation or therapeutic administration schedule	Thrombotic/fibrinolytic-related indicators, toxicity (liver/kidney, etc.), off-target effects, immune response	Securing a therapeutic window where efficacy is maintained within safety limits
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C1 (in vivo efficacy replication) * C3 (enhancement of epithelial aging axis mechanisms) → Validation of epithelial–fibroblast crosstalk → Summary of the stepwise validation sequence for safety and therapeutic window assessment, along with models/interventions/key readouts/success criteria/risks for each stage.

5 Limitations

This work relies on literature-based evidence structuring and automated evaluation; therefore, conclusions may be influenced by coverage and bias of the collected literature. This study relies on literature-based structuring and automated evaluation, so the scope and quality of evidence may be influenced by the coverage and bias of the collected literature. To mitigate this, we plan to expand the search scope (diversifying keywords/databases) and cross-validate key conclusions using independent data (public omics or subsequent experiments) (Table 3).

Additionally, since C1/C2/C3 classifications are rule-based, there is potential for bias depending on rule design. In uncertain cases, they are conservatively classified as UNCERTAIN, but future stabilization will be pursued through a human-in-the-loop approach involving human review and model/data expansion (Table 2).

Finally, the network integration presented in this paper constitutes a conceptual summary connecting literature evidence. Large-scale quantitative validation under the same criteria (e.g., meta-analysis level) and enhancing reproducibility (C2) across diverse models remain as future tasks (Table 3).

Table 3: Summary of limitations and plans for improvement in this study

Limitation	Why it matters	Mitigation / Next step
Literature-based (no new wet-lab studies)	Scope and quality of evidence dependent on collected literature	Expand additional literature and validate with independent data/experiments (animal/cell/co-culture)
Rule-based automated classification (C1/C2/C3)	Potential bias based on baseline design (weighting/thresholding)	Stabilization through baseline calibration; human-in-the-loop review of samples; model/data extension
Conceptual limitations of network centrality analysis	Limited to quantitative superiority conclusions (effect size/meta-analysis)	Integration at the level of quantitative meta-analysis; network validation using publicly available omics/clinical data; strengthening reproducibility (C2) across diverse models

We summarized key limitations, such as potential biases in literature-based approaches and rule-based automated judgments, and quantitative verification limitations in network integration analysis. We also outlined the impact of each limitation on result interpretation and subsequent plans to mitigate them (data/literature expansion, human-in-the-loop review, quantitative verification, and enhanced reproducibility).

6 Conclusion

This study established a literature-based RAG automated evaluation pipeline to structure and assess candidate targets in the aging–fibrosis axis of IPF based on C1 (efficacy), C2 (reproducibility), and C3 (mechanism connectivity) criteria, identifying PAI-1 (SERPINE1) as a priority validation candidate. PAI-1 is identified as a candidate intervention point where the aging-related axis and the fibrosis execution axis intersect. This conclusion is supported by evidence from in vivo studies showing reduced collagen quantification (hydroxyproline) following PAI-1 inhibition (TM5275) (C1) and decreased p53/p21 protein expression upon PAI-1 inhibition (siRNA) (C3). Furthermore, by proposing PAI-1 target peptide candidates through strategy.

7 References

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References

[1] References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to small (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

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A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**

AI Co-Scientist Challenge Korea Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

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1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The paper explicitly describes the development of a literature-based RAG pipeline and its application to prioritize PAI-1 as a target, matching the presented results.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 (Limitations) specifically discusses literature-based biases, rule-based classification constraints, and the need for experimental validation.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This study focuses on an empirical computational pipeline and target

screening and does not include new mathematical theorems or formal proofs.

4. **Experimental Result Reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: This The Methods section (2.1 and 2.2) details the five-stage RAG pipeline, keywords used for collection, and the specific C1/C2/C3 evaluation criteria.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Due to the contest regulations and pending internal review, the code and processed datasets are not publicly released but can be provided upon reasonable request.

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 2 and Table 1 describe the normalization schema, search keywords, and the specific structural templates (e.g., PDB: 1DB2) used for docking.

7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The study utilizes a literature-based RAG pipeline and standard docking software, which are described in the Methods and Section 3.5.

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The study utilizes a literature-based RAG pipeline and standard docking software, which are described in the Methods and Section 3.5.

9. **Code Of Ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research utilizes publicly available literature and structural data, and does not involve unethical data scraping or human/animal rights violations.

10. **Broader Impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion (Section 4) and Conclusion (Section 6) highlight the therapeutic potential for IPF patients while noting safety risks related to PAI-1's physiological functions.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The study focuses on targeted drug discovery for fibrosis and does not involve high-risk generative models or sensitive personal data.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Original studies providing the experimental data (e.g., Omori et al., 2016) and structural templates are cited throughout the text and figures.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The literature-based RAG automated evaluation pipeline and the newly proposed peptide candidates are documented in Sections 2 and 3.5.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This study does not involve crowdsourcing or research conducted directly on human subjects.

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: Since the study is based on existing literature data and computational modeling, IRB approval was not required.